

Лабораторная работа 4

Решающие деревья

1. Запустить на своем компьютере и разобрать пример из раздела «Использование Scikit-Learn. Учимся строить дерево решений» (с.298) книги «Грокаем машинное обучение». Блокнот Graphical_example.ipynb с предлагаемым решением находится по адресу:

[manning/Chapter_09_Decision_Trees/Graphical_example.ipynb at master · luisguiserrano/manning](https://manning.github.io/Chapter_09_Decision_Trees/Graphical_example.ipynb)

Для запуска примера на вашем компьютере необходимо будет доустановить некоторые библиотеки.

Еще необходимо в методе display_tree() заменить импорт

```
from sklearn.externals.six import StringIO на  
from io import StringIO
```

2. Запустить на своем компьютере и разобрать пример из раздела «Реальное применение. Моделирование приема студентов с помощью Scikit-Learn» (с. 300) той же книги. Блокнот University_admissions.ipynb:

[manning/Chapter_09_Decision_Trees/University_Admissions.ipynb at master · luisguiserrano/manning](https://manning.github.io/Chapter_09_Decision_Trees/University_Admissions.ipynb)

Соответствующий тренировочный набор данных можно найти по адресу:

www.kaggle.com/datasets/mdraiyanhossain/admission

Для загрузки тренировочных данных запускайте в блокноте ячейку с пометкой: # IMPORTANT: ONLY RUN THIS CELL IF YOU HAVE CLONED THE REPO

Обратите внимание, что название колонки 'Chance of Admit ' в файле, полученном по последнему URL, заканчивается пробелом.

3. Индивидуальное задание (часть 1). Аналогичным образом построить решающие деревья для данных, соответствующих вашему варианту, приведенному ниже.

4. Индивидуальное задание (часть 2). Построить различные виды решающих деревьев (как в пункте 2) для наборов данных, указанных в таблице 1. Предварительно разделите данные на обучающую и тестировочную выборки в соотношении 80:20.

Для разных способов построения решающего дерева найти матрицу ошибок классификации (Confusion matrix) и метрики качества классификации (accuracy, precision, recall). Найти самое удачное решающее дерево.

Индивидуальные задания для части 1

Вариант 1

data_x = [(207, 242), (378, 73), (392, 170), (108, 222), (311, 296), (260, 233), (278, 377), (318, 344), (161, 181), (205, 293), (301, 131), (318, 100), (376, 47), (105, 175), (203, 169), (139, 104), (79, 248), (300, 204), (116, 77), (344, 50), (179, 154), (310, 55), (206, 338), (226, 261), (358, 114), (331, 160), (297, 57), (281, 339), (156, 127), (327, 233), (257, 51), (261, 113), (237, 313), (296, 107), (275, 270), (372, 130), (126, 133), (272, 322), (97, 121), (405, 148), (323, 120), (342, 78), (338, 322), (193, 301), (361, 174), (368, 242), (301, 162), (383, 109), (406, 64), (281, 163), (372, 163), (80, 194), (142, 182), (232, 353), (344, 136), (407, 101), (304, 246), (289, 80), (145, 212), (261, 331), (182, 276)]
data_y = [2, 3, 3, 1, 2, 2, 2, 2, 1, 2, 3, 3, 3, 1, 1, 1, 1, 2, 1, 3, 1, 3, 2, 2, 3, 3, 3, 2, 1, 2, 3, 3, 2, 3, 2, 3, 1, 2, 1, 3, 3, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 3, 1, 1, 2, 3, 3, 2, 3, 1, 2, 2]

Вариант 2

data_x = [(445, 310), (234, 123), (280, 265), (130, 241), (154, 329), (137, 154), (157, 281), (354, 332), (401, 288), (289, 280), (227, 388), (230, 105), (101, 338), (164, 70), (267, 86), (59, 171), (121, 376), (114, 114), (123, 61), (95, 129), (128, 162), (85, 79), (270, 67), (178, 116), (149, 187), (343, 196), (281, 251), (374, 309), (327, 114), (222, 87), (167, 401), (257, 118), (315, 326), (439, 350), (199, 131), (132, 79), (246, 343), (414, 342), (221, 357), (341, 151), (59, 125), (207, 401), (113, 278), (312, 269), (123, 315), (386, 361), (144, 118), (116, 136), (145, 294), (123, 273), (416, 287), (158, 314), (320, 74), (173, 353), (184, 183), (108, 314), (229, 64), (294, 251), (76, 162), (295, 148), (152, 259), (131, 338), (384, 327), (151, 359), (205, 109), (192, 199), (409, 307), (276, 128), (300, 103), (105, 159), (159, 118), (122, 125), (327, 304), (131, 100), (325, 287), (114, 181), (311, 174), (170, 370), (189, 354), (145, 377), (350, 315), (177, 162), (164, 169), (403, 367), (391, 350), (277, 161), (191, 91), (156, 142), (332, 290), (171, 89), (95, 107), (220, 334), (81, 87), (171, 137), (210, 167), (353, 349), (287, 310), (441, 335), (255, 261), (303, 291), (333, 214), (174, 387), (104, 251), (293, 200), (371, 295), (437, 293), (181, 64), (424, 366), (418, 314), (199, 375), (197, 135), (254, 369)]
data_y = [3, 1, 3, 2, 2, 1, 2, 3, 3, 3, 2, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 3, 3, 1, 1, 2, 1, 3, 3, 1, 1, 2, 3, 2, 1, 1, 2, 2, 3, 2, 3, 1, 1, 2, 2, 3, 2, 1, 2, 1, 2, 1, 3, 1, 1, 2, 2, 3, 2, 1, 1, 3, 1, 1, 1, 1, 1, 3, 1, 3, 1, 1, 2, 2, 2, 3, 1, 1, 3, 3, 1, 1, 1, 3, 1, 1, 2, 1, 1, 1, 3, 3, 3, 3, 3, 1, 2, 2, 1, 3, 3, 1, 3, 3, 2, 1, 2]

Вариант 3

data_x = [(435, 428), (417, 166), (120, 290), (361, 309), (39, 209), (125, 180), (119, 139), (294, 319), (315, 320), (131, 304), (419, 254), (317, 181), (422, 127), (432, 252), (403, 228), (346, 460), (303, 241), (381, 154), (460, 445), (148, 343), (323, 330), (296, 469), (254, 281), (349, 478), (438, 468), (230, 155), (99, 137), (285, 387), (266, 142), (299, 413), (265, 127), (386, 434), (230, 144), (436, 182), (215, 232), (406, 185), (268, 303), (65, 181), (317, 116), (357, 358), (102, 292), (333, 348), (128, 317), (202, 338), (350, 284), (138, 189), (112, 325), (393, 195), (149, 160), (377, 433), (173, 174), (312, 264), (80, 199), (395, 140), (448, 436),

(411, 231), (320, 290), (321, 476), (245, 312), (362, 451), (270, 321), (391, 457), (239, 330), (384, 111), (421, 222), (300, 386), (197, 339), (83, 221), (385, 467), (275, 429), (214, 174), (289, 187), (131, 213), (31, 233), (275, 444), (210, 198), (404, 209), (357, 337), (308, 452), (404, 322), (416, 405), (241, 194), (109, 269), (192, 307), (437, 210), (157, 320), (166, 343), (341, 316), (375, 315), (251, 446), (238, 290), (147, 233), (265, 426), (385, 354), (184, 239), (417, 449), (404, 338), (395, 166), (213, 314), (215, 300), (42, 187), (347, 250), (134, 339), (382, 184), (151, 308), (68, 235), (410, 153), (280, 471), (452, 455), (170, 308), (316, 232), (452, 405), (260, 196), (323, 438), (216, 163), (348, 269), (418, 425), (277, 265), (460, 415), (372, 412), (279, 405), (336, 421), (332, 232), (361, 178), (183, 320), (203, 209), (365, 434), (454, 159), (469, 462), (352, 297), (151, 205), (252, 463), (304, 157), (92, 189), (252, 165), (330, 273), (311, 382), (292, 288), (388, 315), (256, 326), (302, 136), (106, 311), (175, 219), (413, 460), (396, 476), (304, 442), (348, 141), (249, 211), (278, 161), (416, 197), (395, 406), (230, 183), (70, 161), (100, 168), (94, 207), (317, 407), (111, 198), (291, 357), (327, 455), (337, 147), (375, 331), (45, 217), (175, 189), (347, 160)]

data_y = [3, 1, 2, 2, 1, 1, 1, 2, 2, 2, 1, 1, 1, 1, 1, 3, 2, 1, 3, 2, 2, 3, 2, 3, 3, 1, 1, 3, 1, 3, 1, 3, 1, 1, 1, 1, 2, 1, 1, 2, 2, 2, 2, 2, 2, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 2, 3, 2, 3, 2, 3, 2, 1, 1, 3, 2, 1, 3, 3, 1, 1, 1, 1, 3, 1, 1, 2, 3, 2, 3, 1, 2, 2, 1, 2, 2, 2, 2, 3, 2, 1, 3, 2, 1, 3, 2, 1, 2, 2, 1, 2, 2, 1, 2, 1, 1, 3, 3, 2, 2, 3, 1, 3, 1, 2, 3, 2, 3, 3, 3, 3, 2, 1, 2, 1, 3, 1, 3, 2, 1, 3, 1, 1, 1, 2, 3, 2, 2, 2, 1, 2, 1, 3, 3, 3, 1, 1, 1, 1, 3, 1, 1, 1, 3, 1, 3, 3, 1, 2, 1, 1, 1]

Вариант 4

data_x = [(73, 165), (177, 418), (93, 102), (194, 203), (130, 330), (378, 308), (370, 447), (212, 305), (316, 399), (141, 371), (137, 177), (217, 192), (240, 305), (334, 107), (286, 132), (313, 311), (343, 189), (205, 431), (360, 344), (327, 371), (415, 322), (178, 305), (331, 230), (416, 366), (68, 126), (385, 166), (246, 333), (57, 149), (378, 182), (156, 235), (91, 221), (111, 155), (357, 99), (227, 164), (101, 167), (406, 305), (359, 343), (103, 128), (84, 243), (221, 213), (404, 412), (244, 363), (103, 202), (179, 214), (344, 174), (205, 124), (343, 125), (246, 451), (349, 130), (134, 223), (396, 434), (293, 194), (314, 377), (383, 123), (152, 201), (56, 187), (218, 173), (61, 208), (149, 148), (408, 163), (338, 258), (361, 311), (306, 116), (185, 169), (128, 131), (201, 263), (383, 244), (372, 74), (185, 338), (121, 81), (392, 343), (374, 385), (278, 153), (324, 454), (199, 146), (149, 109), (281, 249), (206, 347), (193, 373), (294, 307), (248, 239), (266, 218), (351, 215), (336, 73), (317, 165), (279, 454), (275, 408), (306, 203), (167, 190), (264, 179), (395, 118), (317, 91), (322, 343), (170, 132), (266, 144), (267, 387), (110, 257), (354, 419), (175, 92), (202, 231), (401, 219), (323, 257), (226, 141), (119, 201), (315, 154), (187, 234), (272, 348), (175, 392), (314, 215), (255, 285), (227, 404), (297, 233), (149, 265)]

data_y = [1, 3, 1, 1, 3, 3, 3, 3, 3, 3, 1, 2, 3, 2, 2, 3, 2, 3, 3, 3, 3, 3, 2, 3, 1, 2, 3, 1, 2, 1, 1, 1, 2, 2, 1, 3, 3, 1, 1, 1, 3, 3, 1, 1, 2, 1, 2, 3, 2, 1, 3, 1, 3, 2, 1, 1, 1, 1, 1, 2, 2, 3,

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Вариант 5

data_x = [(236, 342), (88, 307), (188, 306), (359, 385), (380, 206), (335, 224), (201, 185), (124, 130), (102, 401), (390, 232), (210, 123), (72, 180), (440, 166), (69, 234), (239, 245), (185, 217), (136, 312), (424, 87), (136, 239), (142, 213), (339, 143), (214, 363), (129, 302), (200, 226), (250, 240), (214, 235), (391, 410), (441, 378), (206, 142), (360, 71), (199, 284), (288, 229), (142, 158), (260, 210), (85, 174), (235, 87), (280, 173), (287, 310), (82, 258), (335, 191), (260, 269), (78, 122), (252, 142), (255, 332), (123, 357), (134, 355), (390, 360), (206, 379), (142, 100), (212, 391), (229, 274), (241, 294), (115, 304), (212, 163), (455, 301), (218, 314), (390, 286), (270, 374), (256, 351), (210, 292), (120, 253), (92, 376), (49, 286), (109, 183), (55, 381), (235, 160), (236, 367), (251, 283), (193, 329), (429, 265), (103, 272), (452, 134), (196, 203), (79, 367), (98, 133), (78, 291), (452, 251), (219, 195), (393, 164), (421, 229), (232, 188), (289, 268), (98, 347), (356, 332), (205, 354), (353, 278), (247, 224), (229, 217), (252, 180), (205, 255), (185, 157), (95, 80), (414, 310), (69, 331), (54, 224), (269, 131), (257, 390), (205, 78), (192, 247), (448, 196), (49, 342), (114, 233), (195, 285), (270, 108), (266, 299), (398, 121), (200, 320), (71, 404), (129, 397), (433, 383), (123, 165), (228, 117), (139, 267), (381, 124), (96, 224), (109, 109), (188, 176), (124, 192)]

data_y = [2, 1, 2, 3, 2, 3, 2, 1, 1, 3, 2, 1, 3, 1, 2, 2, 1, 3, 1, 1, 3, 2, 2, 2, 2, 1, 3, 3, 2, 3, 2, 2, 1, 2, 1, 2, 2, 2, 1, 2, 2, 1, 2, 2, 1, 1, 3, 2, 1, 2, 2, 2, 1, 2, 3, 2, 3, 2, 2, 2, 2, 1, 1, 1, 1, 2, 2, 2, 1, 3, 1, 3, 2, 1, 1, 1, 2, 1, 3, 2, 2, 2, 1, 3, 2, 3, 2, 2, 2, 2, 2, 1, 3, 1, 1, 2, 2, 2, 2, 3, 1, 1, 1, 2, 2, 3, 2, 1, 1, 3, 1, 2, 1, 3, 1, 1, 1, 2]

Вариант 6

data_x = [(304, 158), (167, 138), (166, 255), (389, 124), (302, 104), (403, 187), (84, 243), (308, 433), (173, 112), (104, 352), (409, 295), (77, 201), (241, 32), (459, 252), (144, 353), (378, 173), (462, 185), (259, 165), (388, 359), (192, 65), (443, 116), (323, 215), (128, 132), (270, 174), (347, 173), (329, 123), (157, 303), (369, 293), (335, 369), (294, 372), (82, 90), (270, 304), (204, 283), (266, 182), (287, 32), (220, 199), (282, 109), (177, 421), (466, 345), (228, 324), (219, 270), (257, 183), (319, 245), (97, 103), (141, 185), (184, 346), (362, 416), (313, 302), (234, 373), (408, 89), (200, 106), (228, 384), (446, 304), (466, 377), (303, 205), (215, 44), (330, 254), (193, 218), (130, 399), (388, 226), (72, 124), (214, 231), (452, 93), (430, 150), (114, 216), (145, 95), (184, 385), (297, 279), (303, 145), (430, 184), (462, 154), (113, 72), (271, 334), (360, 282), (403, 324), (365, 434), (345, 321), (313, 304), (323, 75), (416, 146), (395, 393), (118, 265), (92, 301), (150, 139), (379, 325), (344, 100), (441, 78), (406, 365), (466, 286), (359, 353), (369, 230), (204, 137), (263, 417), (398, 263), (232, 90), (254, 133), (242, 295), (363, 382), (236, 124), (177, 179), (297, 368), (91, 155), (261, 111), (327, 339), (363, 147),

199), (315, 67), (89, 382), (366, 279), (316, 309), (157, 175), (295, 117), (389, 182), (268, 421), (135, 213), (195, 298), (156, 146), (421, 314), (119, 44), (253, 229), (413, 256), (277, 97), (385, 432), (83, 266), (100, 84), (394, 283), (106, 333), (328, 388), (111, 446), (427, 132), (129, 146), (279, 343), (190, 227), (115, 134), (130, 329), (294, 285), (364, 382), (192, 275), (312, 200), (156, 127), (239, 311), (147, 82), (215, 255), (85, 95), (244, 264), (307, 231), (90, 313), (312, 267), (398, 393), (424, 391), (428, 95), (151, 362)]
data_y = [1, 2, 1, 2, 2, 3, 1, 2, 1, 1, 3, 2, 1, 3, 3, 1, 2, 2, 2, 3, 3, 1, 3, 2, 2, 2, 2, 1, 1, 2, 1, 1, 1, 2, 1, 1, 2, 2, 2, 1, 2, 2, 1, 3, 1, 2, 3, 3, 1, 3, 3, 2, 1, 2, 1, 2, 1, 1, 3, 3, 1, 2, 2, 2, 1, 2, 3, 2, 3, 1, 2, 1, 2, 2, 3, 2, 2, 2, 1, 1, 1, 1, 2, 3, 2, 2, 1, 3, 2, 1, 2, 3, 2, 1, 2, 1, 3, 1, 2, 3, 2, 3, 1, 1, 3, 1, 2, 1, 3, 1, 2, 2, 1, 1, 2, 3, 1, 2, 1, 2, 1, 1, 1, 2, 2, 1, 2, 3, 3, 3, 1]

Вариант 9

data_x = [(217, 328), (290, 299), (78, 218), (93, 253), (237, 275), (298, 113), (248, 368), (337, 133), (283, 148), (439, 320), (167, 62), (273, 123), (177, 420), (53, 322), (137, 132), (203, 354), (234, 416), (306, 72), (238, 271), (404, 218), (163, 350), (176, 285), (183, 465), (301, 267), (227, 76), (311, 175), (289, 194), (440, 142), (86, 434), (462, 254), (228, 198), (70, 178), (132, 277), (173, 175), (258, 104), (348, 297), (396, 349), (120, 464), (130, 57), (63, 383), (304, 400), (332, 204), (102, 84), (274, 313), (289, 347), (132, 180), (204, 221), (127, 404), (244, 162), (357, 251), (398, 55), (255, 315), (305, 362), (382, 113), (78, 122), (417, 293), (349, 362), (296, 435), (372, 175), (319, 306), (101, 375), (137, 212), (139, 322), (247, 249), (170, 244), (88, 267), (270, 255), (457, 197), (179, 97), (272, 282), (243, 212), (216, 126), (97, 324), (233, 463)]
data_y = [3, 3, 1, 1, 1, 2, 2, 2, 1, 2, 1, 2, 2, 3, 1, 3, 3, 1, 2, 2, 3, 1, 3, 2, 1, 2, 1, 2, 3, 2, 1, 1, 1, 1, 1, 2, 2, 3, 1, 3, 3, 2, 1, 2, 2, 1, 1, 3, 2, 2, 1, 2, 3, 2, 1, 2, 2, 3, 2, 2, 3, 1, 3, 3, 3, 3, 1, 2, 1, 3, 2, 1, 3, 3]

Вариант 10

data_x = [(227, 326), (335, 429), (357, 340), (351, 79), (118, 156), (115, 187), (312, 343), (128, 420), (201, 119), (118, 69), (147, 245), (91, 397), (264, 295), (118, 373), (169, 424), (265, 66), (97, 433), (281, 350), (86, 185), (273, 121), (185, 443), (121, 449), (171, 303), (342, 307), (340, 59), (351, 184), (257, 454), (89, 364), (320, 257), (226, 126), (220, 150), (356, 449), (91, 321), (256, 183), (169, 264), (252, 112), (327, 361), (225, 410), (91, 297), (220, 234), (122, 287), (351, 113), (361, 305), (215, 72), (369, 270), (261, 353), (90, 73), (269, 228), (398, 416), (298, 385), (263, 427), (296, 456), (149, 449), (131, 335), (298, 408), (300, 287), (330, 411), (178, 380), (181, 445), (282, 317), (308, 315), (216, 266), (178, 349), (315, 98), (341, 245), (369, 238), (291, 154), (196, 183), (192, 158), (156, 150), (228, 357), (228, 436), (165, 70), (278, 257), (394, 376), (381, 449), (365, 390), (153, 378), (351, 149), (379, 351), (117, 111), (220, 292), (96, 244), (83, 143), (148, 308), (312, 289), (344, 376), (259, 413), (88, 109), (321, 220), (161, 184),

(304, 185), (315, 143), (250, 258), (245, 378), (176, 412), (256, 154), (305, 69), (330, 294), (151, 111), (166, 245)]

data_y = [2, 3, 3, 1, 1, 1, 3, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, 2, 1, 1, 3, 1, 2, 1, 1, 3, 1, 1, 1, 3, 3, 1, 2, 1, 1, 2, 1, 2, 3, 1, 2, 3, 3, 3, 3, 1, 1, 3, 2, 3, 1, 1, 2, 2, 2, 1, 1, 2, 2, 1, 1, 1, 1, 3, 3, 1, 2, 3, 3, 3, 1, 1, 3, 1, 2, 1, 1, 1, 2, 3, 3, 1, 2, 1, 1, 1]

Таблица 1

Вариант	Набор данных
1	BankNote_Authentication.csv
2	car_evaluation.csv
3	diabetes.csv
4	haberman.csv
5	iris_binary.csv